InvoiceVault

Programmable invoice-backed credit infrastructure using IOTA notarization and Move smart contracts to create auditable, on-chain lifecycle management of SME financing.

Tokenise · Fund · Settle · Track

CONTEXT & TREND

Three structural forces are making receivables programmable — across all of Europe

€340B+

annual SME liquidity gap
across Europe

ECB Survey on SME Access to Finance

60+

days avg. payment period
EU B2B / G2B transactions

EU Payment Observatory 2024

52%

of EU companies report
late payment impact on operations

European Commission / Intrum EPR

THREE FORCES CONVERGING

Late Payments Persist
~74 days/yr chasing (Intrum)

→

E-Invoicing Goes EU
ViDA mandate + SdI at 2.4bn/yr

→

Tokenisation Matures
BIS/FSB frameworks defined

= **InvoiceVault**

PROBLEM & OPPORTUNITY

SMEs involuntarily finance their customers' working capital

The Pain (Europe-wide)

- 1 in 4 EU bankruptcies linked to invoices not paid on time (European Commission)
- DSO averages 55-65 days across EU; 80+ days in Southern Europe — cash trapped in receivables
- SMEs lack bargaining power: larger counterparties dictate payment terms
- Traditional factoring: onboarding friction, opacity, and manual underwriting that doesn't scale to small tickets

The Opportunity

- EU ViDA mandate will make e-invoicing universal — machine-readable data at continental scale
- Standardised invoice data makes verification automatable and underwriting costs collapsible
- On-chain lifecycle creates measurable, portable risk signals — replacing PDF-based trust
- Not "banks are bad" — the data infrastructure to serve small tickets at scale simply didn't exist until now

EUROPEAN LATE PAYMENT LANDSCAPE

Avg. Days Sales Outstanding by market (Intrum EPR / Euler Hermes)

SOLUTION

Invoice claims become programmable on-chain credit instruments

THIS IS A TYPE OF RECOUSE FACTORING

Invoice-backed short-term credit with recourse to seller. The invoice is the auditable reference instrument — not the collection target. Repayment comes from the issuer at maturity. The platform never takes balance-sheet risk. It's seller-backed short-term liquidity with an immutable on-chain lifecycle.

Start Simple, Scale Later

Faster & simpler go-to-market: No need to run a full collections + legal assignment machine on day 1 → fewer operational/legal frictions, easier pilots.

Keep the Customer Relationship Untouched

Zero disruption to the debtor relationship: The SME keeps the customer relationship untouched (no "factor calling the buyer") → higher adoption for SMEs.

Cleaner underwriting for small tickets

Cleaner underwriting for small tickets: Risk is primarily seller performance (repayment behavior), not per-debtor collection complexity → scalable to long-tail SMEs.
Lower unit cost through automation: Smart-contract states + fee logic reduce manual work → viable economics on smaller invoices.

On-Chain Reputation as the Moat.

On-chain reputation becomes the moat: DEFAULTED / RECOVERED + ratings create a portable, auditable risk history that enables better pricing over time.

VALUE PROPOSITION

Liquidity, transparency, and programmable risk — built for SMEs

Sellers (SMEs)

- Faster cash conversion cycle
- Predictable, transparent fees
- Repayment history = on-chain credit reputation
- Recourse model = seller stays in control

Buyers (Capital Providers)

- Yield via invoice discount (short-duration)
- Real-time portfolio visibility on-chain
- State-based risk signals, not opaque data
- Diversifiable across issuers and tenors

Ecosystem

- Standardised receivables financing primitives
- Machine-readable compliance data (MiCA-ready)
- Integration with EU e-invoicing (ViDA/SdI)
- Audit-ready infrastructure from day one

MVP DELIVERED — END-TO-END ON-CHAIN

- ✔ Full lifecycle: create claim → list → fund → repay — wallet-signed, real transaction digests on IOTA
- ✔ Demo-mode: accelerated maturity, default/recovery simulation, rating continuity, self-funding guards, fee logic

HOW IT WORKS

Verifiable lifecycle: from invoice claim to repayment — with recourse to issuer

01
Create Claim

Seller anchors invoice hash + metadata on-chain via IOTA Notarization

Implemented

→

02
List

Claim published on marketplace: discount, due date, terms — all transparent

Implemented

→

03
Fund

Buyer funds at discount; 0.75% fee applied; claim state → Funded

Implemented

→

04
Settle

Issuer repays at maturity (recourse); portfolio and rating update on-chain

Implemented

Recourse Model:
Repayment obligation stays with the seller/issuer. The invoice is the reference instrument, not the collection target. Investor risk = issuer credit risk.

- **On-chain default tracking:** after maturity the holder can mark the claim **Defaulted**
- **Recourse stays with issuer:** the issuer remains liable and can still settle via **recovery**
- **MVP demo fee:** recovery = **principal + 8% default fee** (paid to the holder)
- **Future releases:** calibrate fee with **time-based accrual (days past due)** + caps / risk tiers

FIVE-LAYER ARCHITECTURE

Provenance	Hash + SdI integration
Credit	Move contract lifecycle
Settlement	Funds + fees + escrow
Risk	Scorecards + limits
Data/Index	Incremental sync + UI

Built on IOTA (Move language) — resource model prevents asset duplication by design

TARGET CUSTOMERS

SMEs with verified invoicing rails + institutional capital providers

Sellers

SME owner / CFO

- B2B invoices, 45-90 day payment cycles
- Italy first (SdI), then EU via ViDA
- Needs liquidity without restructuring
- Limited power vs. larger counterparties

Buyers

Credit Investor / Fund

- Short-duration yield (30-90 day tenors)
- Alt-credit funds, corporate treasuries
- On-chain risk signals + portfolio view
- Concentrated early for feedback loops

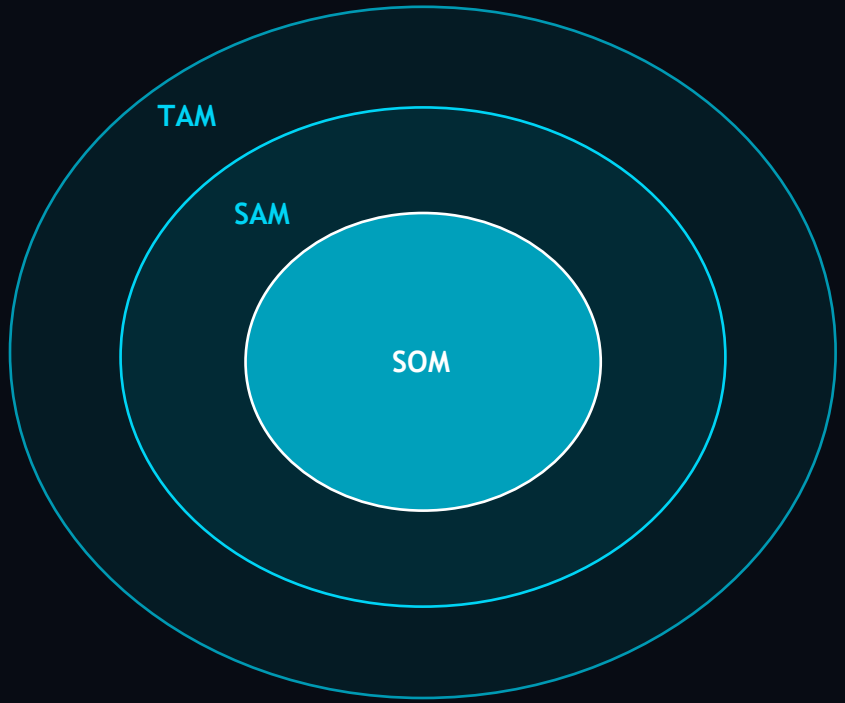
Partners

Integration Enablers

- E-invoicing / ERP connectors (SdI, Peppol)
- Accounting platforms, SME associations
- KYB/KYC/AML providers (compliance rail)
- Credit insurers (target expansion)

MARKET OPPORTUNITY

A multi-trillion euro receivables finance market



TAM · Global Factoring Turnover (2024)

€3.9 Trillion

FCI (Factors Chain International)

SAM · Europe Factoring + Commercial Finance (2024)

€2.1 Trillion

EUF / EU Federation

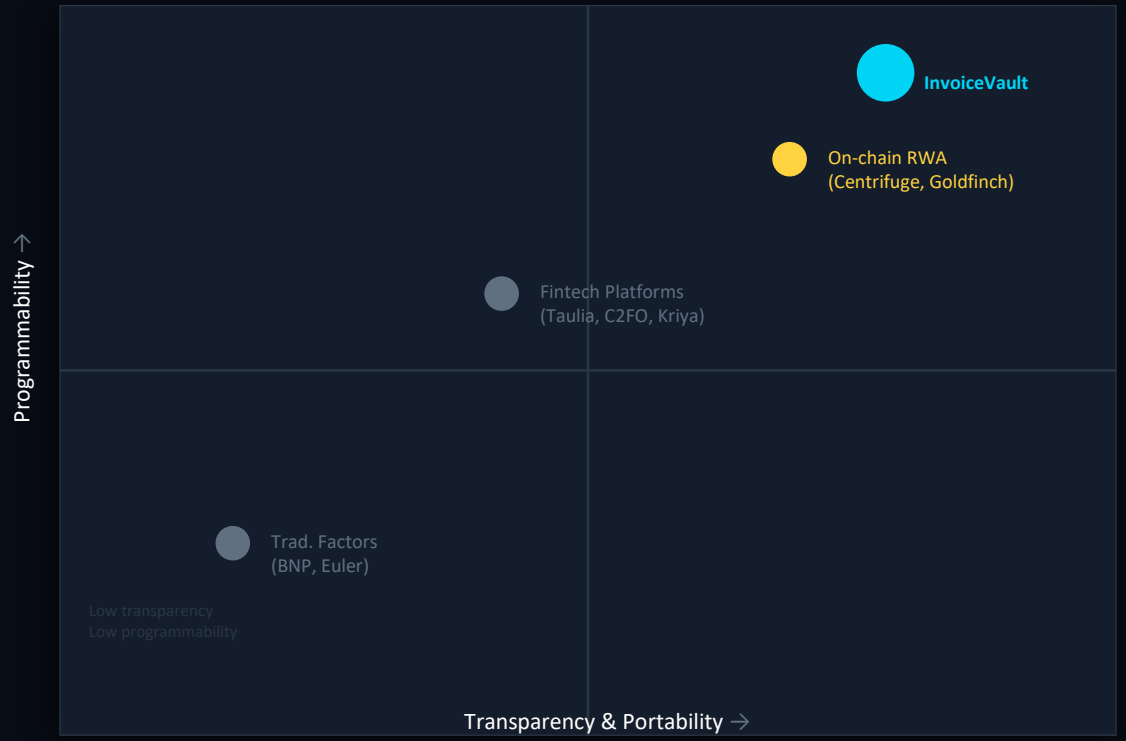
SOM · Beachhead Markets

Italy	**€289B**	SdI live, mass e-invoicing
Spain	**€208B**	2nd largest EU factoring
France	**€403B**	Largest EU factoring market
Germany	**€310B**	ViDA early adopter corridor

*Italy-first beachhead; **5-10%** capture → **€14.5-28.9B** financed/yr*
*@ 0.75% fee → **€110-220M** platform revenue (fee-only, excl. add-ons)*

COMPETITIVE LANDSCAPE

Competing on transparency + programmable lifecycle — not just faster lending



WHY WE WIN

Immutable Audit Trail

Every state transition verifiable on-chain — not in a proprietary dashboard

Programmable Contract Logic

Fees, overdue, recovery, self-funding guards — all in Move smart contracts

Recourse Model Clarity

Issuer liability explicit from mint to maturity — no collection risk confusion

Production-Aligned MVP

Same primitives demo → production; hardening the rails, not rebuilding them

REVENUE MODEL

Transaction-based take-rate with data subscription upside

Primary Fee

Platform fee on successful funding (paid by buyer)

0.75% of funded amount

Target Additions

Analytics / scorecard subscription
Servicing / collections fees
Compliance orchestration (KYB/AML)
API fees for partners
Tiered roadmap pricing (0.50% <=30 d, 0.75% 31-60d, 0.75% > 60d

Why 0.75% Works

- Sits within auction-style benchmark range for 60-day tenors
- Below traditional factoring service fees (0.75–2.5%)
- Transparent: every step traceable on-chain

ILLUSTRATIVE TRANSACTION MECHANICS

Invoice	→	Funded	→	Platform Fee	→	Buyer Outlay	→	Issuer Repays
€10,000		€9,700		€72.75		€9,772.75		€10,000
Face value		3% discount		0.75% of funded		Funded + fee		At maturity (recourse)

Numbers purely illustrative. Platform earns fee for facilitating funding, without balance-sheet risk. Buyer return = €227.25 gross over tenor.

GO-TO-MARKET

MVP proven on-chain → regulated pilot → EU scaling

VALIDATION

NOW ✓

- MVP live (end-to-end lifecycle)
- Wallet signing + tx digests
- Default/recovery demo mode

PILOT

3-6 MO

- Italy issuer-buyer cohort
- KYB/KYC + SdI provenance
- Basic risk limits + ratings

BUILD-OUT

6-12 MO

- Indexer + analytics dashboard
- Smart contract audit
- Collections / overdue engine

EU LAUNCH

12-18 MO

- MiCA/DLT Pilot compliance
- Spain + France expansion
- Peppol + ViDA integration

Key Metrics: Time-to-fund · Repayment performance · Repeat usage rate · Default-to-recovery ratio

FUNDING & USE OF PROCEEDS

Run a regulated pilot and harden the platform for EU scale

THE ASK **€100K – €250K**

First live pilot → measured repayment data → issuer scorecards → regulatory pathway

Product & Engineering
Indexer, e-invoicing integrations (SdI + Peppol), testing automation

35%

Security & Compliance
Smart contract audit, KYB/KYC vendor integration, MiCA / DLT Pilot legal structuring

25%

GTM & Partnerships
Pilot issuer/buyer acquisition, channel partnerships (accounting, e-invoicing, SME associations)

25%

Operations
Infrastructure hardening, monitoring, incident response (ENISA security framework)

15%

Regulatory context: MiCA (crypto-asset framework) + EU DLT Pilot Regime (since March 2023) — phased compliance, not regulatory arbitrage.

TEAM

Fintech domain × protocol engineering × compliance

CEO / BD

SME + fund partnerships
Regulatory strategy
Pilot execution

CTO / Protocol

Move smart contracts
Indexing & security
System architecture

Full-Stack / Product

Next.js UX + wallet integration
Analytics dashboards
E-invoicing connectors

TARGET ADVISORS

Receivables Finance · EU Compliance / KYC · Credit Risk · IOTA Ecosystem

Domain experts to be onboarded during pilot phase

Execution Proof: MVP shipped end-to-end on IOTA — wallet signatures, on-chain state transitions, fee logic, default/recovery lifecycle, and self-funding guards all validated.

Design Principle:

InvoiceVault tokens are the on-chain control plane for a claim lifecycle. Legal enforceability requires off-chain legal wrappers — exactly what the production roadmap hardens. This is invoice-backed short-term credit with recourse to seller.

APPENDIX | MVP LEGAL-RISK MATRIX

Same MVP, but with payments kept on-chain via an authorised euro-EMT settlement design

LAUNCH-CRITICAL MVP FEATURES

RED = blocker | YEL = controls | GRN = safe

MVP FEATURE	R/Y/G	ITALY FIX	EU FIX
On-chain invoice hash + metadata	YEL	Hash only on-chain SdI XML, debtor data + assignment docs off-chain	GDPR-by-design DPIA / role map; avoid personal data on-chain
Marketplace listing + buyer funding	RED	Bank / Art.106 partner or licensed broker mandate Financing + receivables docs off-chain	No public offer until classification memo is done Choose MiCA vs securities path up front
On-chain EUR EMT payments + fees	YEL	CASP + PSP/IMEL/bank licence or partnership No platform treasury / omnibus-wallet risk	TFR / Travel Rule + self-hosted-wallet controls PSD2 can be triggered by EMT transfer / custodial wallets
Seller recourse + default / recovery	YEL	Signed recourse undertaking + payout waterfall mirrored in contracts Manual override / dispute hold	Default notices + enforcement steps explicit Cross-border recovery playbook before rollout
On-chain ratings / scorecards	YEL	Human override + model governance Do not market as consumer-credit scoring	GDPR controls for automated decisions Explainability + challenge process
Own settlement token + EU launch	RED	Do not issue proprietary payment token in MVP Use third-party EUR EMT instead	EMT issuance reserved to EU bank / EMI If security / debt instrument, use securities / DLT path
Audit trail + KYB/KYC/AML	GRN	Whitelisted wallets, sanctions + source-of-funds Signer evidence, immutable logs, versioned terms	Record retention + incident response Keep TFR data across each wallet transfer

BEST PILOT SHAPE Closed Italy B2B cohort + licensed funding/servicing partner + third-party EUR EMT rail + whitelisted wallets + off-chain receivables docs